

Mao Li

COMPUTATIONAL SOCIAL SCIENTIST · SURVEY METHODOLOGIST

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Education

University of Michigan

PH.D. IN SURVEY AND DATA SCIENCE & SCIENTIFIC COMPUTING (DUAL DEGREE)

Ann Arbor, MI

Present

University of Michigan

M.S. IN SURVEY METHODOLOGY

Ann Arbor, MI

April 2021

Nanjing University

BACHELOR OF LAW IN POLITICAL SCIENCE AND ADMINISTRATIVE SCIENCE

Nanjing, China

July 2019

Experience

Graduate Student Research Assistant

University of Michigan, Ann Arbor, MI

INSTITUTE FOR SOCIAL RESEARCH

Sep 2021 - Present

- Developed and applied *topic modeling algorithms* and *text summarization methods* to categorize 10M tweets about the Census Bureau and Census 2020.
- Designed a digital tool (Social Media Browser) to help Census employees analyze and interpret public opinion.
- Utilized LLMs for stance detection using parameter-efficient fine-tuning, enhancing public opinion insights for decision-making.

Project Lead

University of Michigan, Ann Arbor, MI

EMERGING SOCIAL MEDIA PLATFORMS RESEARCH

Nov 2024 - Present

- Built a framework to analyze information cascade patterns on Bluesky and TruthSocial.
- Modeled repost/reply interaction graphs and conducted *motif analysis*.
- Quantified cascades using metrics like structural virality and depth.

Graduate Student Research Assistant

University of Michigan, Ann Arbor, MI

NATIONAL NEIGHBORHOOD DATA ARCHIVE (NANDA)

Sep 2019 - Sep 2021

- Created U.S. neighborhood datasets across spatial scales (e.g., health, socioeconomic, education).
- Used Python, R, ArcGIS, and scraping to integrate data from diverse sources.

Graduate Student Research Assistant

University of Michigan, Ann Arbor, MI

NEIGHBORHOODS & COGNITIVE RESILIENCE PROJECT

Sep 2019 - Sep 2021

- Supported analysis and manuscript writing on neighborhoods' effects on cognition in aging Americans.
- Collaborated with interdisciplinary teams in public health, sociology, and computer science.

Technical Skills

Languages & Tools

Python, C++, R, Spark, Hadoop, Qualtrics,
ArcGIS, MySQL, MongoDB

Data Collection

Survey, Interviews, Web Scraping

Spoken Languages

English, Chinese

Honors

- Janet A. Harkness Student Paper Award (WAPOR)
- Dow Distinguished Award (University of Michigan SEAS)

Projects

Twitter Browser Web App

🔗 [tanukinekolee/Tweet_browser](#)

LEAD DEVELOPER

- Leading the development of a user-friendly web app to summarize Twitter data for non-technical users.
- Implemented advanced summarization algorithms for clear and concise insights.

Publications

Partitioned Realities: How Next Generation Social Media Platforms Shape Competing Narratives of ‘Truth’ ICWSM 2025 Workshop on Next-Gen and Alternative Social Media (Peer-reviewed Workshop Paper)

LI, M. & CHEN, X. 2025

Tracking Census Online Self-Completion Using Twitter Posts Social Science Computer Review

LI, M., & CONRAD, F. 2024

Neighborhood Active Aging Infrastructure and Cognitive Health Preventive Medicine

FINLAY ET AL., INCL. LI, M.

Can Neighborhood Social Infrastructure Modify Cognitive Function? Journal of Aging and Health

FINLAY ET AL., INCL. LI, M.

Brainy neighborhood amenities? A mixed-methods study of intellectually-stimulating infrastructure and cognitive function among older Americans Well-being, Space and Society

FINLAY ET AL., INCL. LI, M.

Neighborhood ‘Disamenities’: local barriers and cognitive function among Black and white aging adults BMC Public Health

YU ET AL., INCL. LI, M.

Under-Review Manuscripts

Language & Dialect Effects on Political Trust in Asian Barometer Survey Journal of Survey Statistics and Methodology

LI, M., CONRAD, F., OWENS, V. 2024

Probing on Self-Rated Health in Multilingual Contexts Journal of Survey Statistics and Methodology

LI, M., MORALES, S., NURUMOV, K., LEE, S. 2024

Advancing Annotation of Stance in Social Media Posts arXiv:2406.07483

LI, M., & CONRAD, F.

FastLexRank: Efficient Lexical Ranking for Social Media arXiv:2410.01183

LI, M., CONRAD, F., & GAGNON-BARTSCH, J.